

Descriptive Statistics

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Outline

1. Classify data and check errors or missing values
2. Visualize and explore the data
3. Measure Central tendency
4. Measure Dispersion
5. Outliers

Classify data

- Numerical or Categorical
- Numerical → Continuous or Discrete
- Categorical → Nominal or Ordinal

Error checking

- Categorical data → Category check
- Numerical data → Range check

Missing values

- Check the missing codes [NAs or user defined NA or tagged NA like -9,99]
- Investigate the reason for missing
 - Completely at random
 - At random
 - Not at random

Data visualization and exploring data

- Numerical variables
 - Histogram
 - Boxplot
- Categorical variables
 - Bar graph
 - Pie chart

Measures of centrl tendency

- Numerical variable
 - if symmetric without extreme value → arithmetic mean
 - if positive skewed → geometric mean
 - if outliers pull the mean → trimmed mean
 - Median is more robust to skewed data

Measures of centrl tendency

- Ordinal variable
 - mode or median
 - mean for comparing two group
- Nominal variable → mode
- Arithmetic mean is useful for inferential statistics or total

Measures of dispersion

- Range shows the total spread: minimum and maximum
- Standard deviation shows average spread for the normally distributed data
 - $2SD - \text{mean} + 2SD$ is 95% of data for normal distribution
- Interquartile range show spread of central 50% of data
 - Inter-decile range shows middle 80% of data
- Coefficient of variation CV shows relative variability

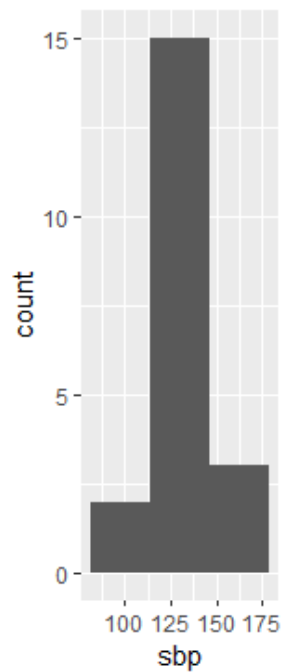
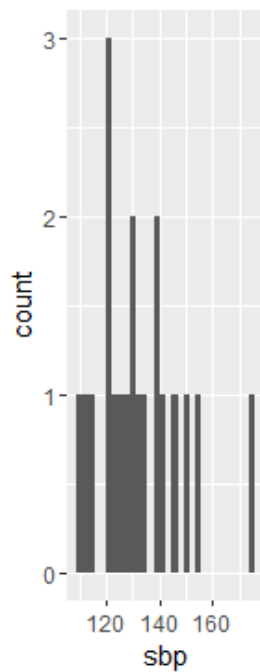
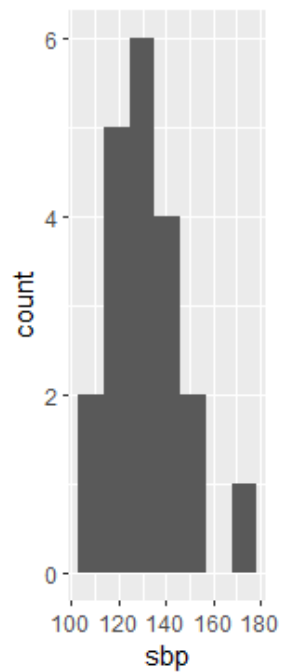
Outlier

- Can be seen in box-plot and stem and leaf plot
- Beyond 1.5 times IQR from the Q1 & Q3
- Needs investigation
 - natural outlier or errors

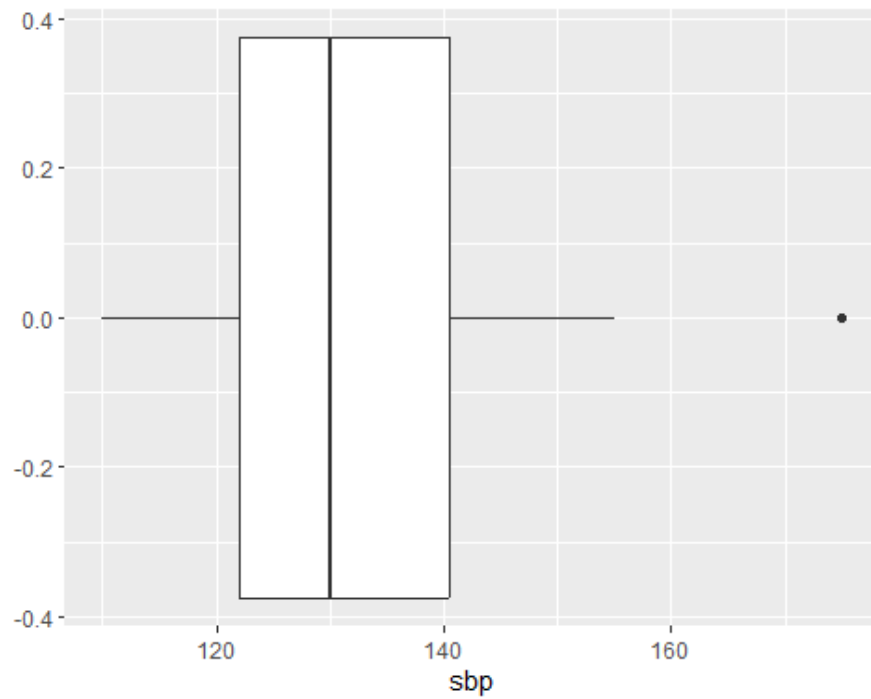
Systolic Blood Pressure (SBP) in 20 Patients (mmHg) after a 4-week trial of Medication X

sbp
110
112
115
120
122
122
124
125
128
130

Histograms



Boxplot



Stem and leaf

The decimal point is 1 digit(s) to the right of the |

10 | 025

12 | 02245800258

14 | 02505

16 | 5

Summary table

Measures	SBP
Mean	132.50
Median	130.00
SD	15.78
IQR	18.50
Q1	122.00
Q3	140.50